

LESSON 6-7 • SESSION 1 • GUIDED NOTES

Factors and Greatest Common Factor

Learning target – I can list the factors of two numbers and find their greatest common factor. · 6.NS.B.4

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
factor <i>factor</i>	A whole number that divides another number with ____ remainder.	1, 2, 4, 8, 16 all divide 16
common factor <i>factor común</i>	A factor that two numbers ____.	4 is a factor of 16 and of 24
greatest common factor (GCF) <i>máximo común divisor</i>	The ____ factor that two numbers share.	GCF of 16 and 24 is 8
prime factorization <i>factorización prima</i>	A number written as a product of ____ numbers.	$24 = 2 \times 2 \times 2 \times 3$

KEY IDEA*complete each sentence*

- List the factors of each number, then circle the factors that appear in ____ lists.
- The GCF tells the greatest number of ____ groups you can make with nothing left over.

WORKED EXAMPLE*fill in the blanks as we solve*

A teacher has 20 pencils and 30 markers. She makes identical kits with nothing left over. What is the greatest number of kits?

Step 1 Factors of 20: 1, 2, 4, 5, ____, 20.

Step 2 Factors of 30: 1, 2, 3, 5, 6, 10, 15, 30. Shared: 1, 2, 5, 10.

Step 3 The GCF is ____, so she can make 10 kits.

Answer Each kit has $20 \div 10 =$ ____ pencils and $30 \div 10 =$ ____ markers.

YOU TRY*use the example as a model*

1. Find the GCF of each pair.

a. 12 and 30 _____

b. 27 and 36 _____

2. A coach has 24 balls and 32 cones. She sets up identical stations with nothing left over. What is the greatest number of stations?

Show your work.

Stations _____

SUM IT UP*in your own words*

To find the GCF of two numbers, I ____ because ____.

LESSON 6-7 • SESSION 2 • GUIDED NOTES

Multiples and Least Common Multiple

Learning target – I can list multiples of two numbers and find their least common multiple. • 6.NS.B.4

WORDS TO KNOW*fill in the missing words*

WORD	WHAT IT MEANS	EXAMPLE
multiple <i>múltiplo</i>	A number you land on when you _____ by another number.	6, 12, 18, 24
common multiple <i>múltiplo común</i>	A number that is a multiple of ____ numbers.	18 is a multiple of 6 and of 9
least common multiple (LCM) <i>mínimo común múltiplo</i>	The _____ number above 0 that is a multiple of both.	LCM of 6 and 9 is 18

KEY IDEA*complete each sentence*

- List multiples of each number and find the _____ number on both lists.
- The product of two numbers is always a common multiple, but not always the _____ one.

WORKED EXAMPLE*fill in the blanks as we solve*

A library van visits every 8 days. An art cart visits every 10 days. Both visit today. In how many days will both visit again?

Step 1 Multiples of 8: 8, 16, 24, 32, ____.

Step 2 Multiples of 10: 10, 20, 30, 40.

Step 3 The first shared multiple is the LCM: ____.

Answer Both visit again in ____ days.

YOU TRY*use the example as a model*

1. Find the LCM of each pair.

a. 4 and 6 _____

b. 5 and 10 _____

2. Plates come in packs of 12. Cups come in packs of 9. What is the fewest of each you can buy to have the same number?

Show your work.

Fewest _____

SUM IT UP*in your own words*

To find the LCM of two numbers, I ____ because ____.
